INVESTIGATING APPROXIMATIONS FOR THE LINEAR AND NON-LINEAR FIRST-ORDER ODE DESCRIBING THE TEMPERATURE OF A PHOTOVOLTAIC CELL

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ABSTRACT. The operating temperature of a photovoltaic (PV) panel is governed by a first-order ordinary differential equation (ODE) that balances absorbed solar irradiance against convective and radiative heat losses. We study two formulations of this equation: a linear model that retains only forced convection, and a nonlinear model that adds a Stefan–Boltzmann radiation term. For the linear case, a closed-form solution is derived and used as a benchmark against which three numerical methods (forward Euler, improved Euler (Heun), and the classical fourth-order Runge–Kutta (RK4) method) are compared over a one-hour simulation with step size $h = 50\,\mathrm{s}$. The nonlinear case, which admits no closed-form solution, is treated by using RK4 as a near-reference trajectory. We subsequently extend the baseline model through five physically motivated enhancements: Beer–Lambert atmospheric attenuation, a diurnal irradiance profile, a temperature-dependent efficiency coupling, a Monte Carlo uncertainty quantification (UQ) study, and a tightly coupled Boltzmann transport equation / Navier–Stokes (BTE–NS) system that couples a dynamic wind-speed boundary-layer ODE to the panel thermal equation. Across all models the nominal panel approaches a steady-state temperature $T_{\mathrm{ss}} \approx 346\,\mathrm{K}$ under standard test conditions, while the more realistic BTE–NS coupled model predicts a diurnal peak that depends sensitively on aerosol loading and boundary-layer wind dynamics. The Monte Carlo study propagates measurement uncertainty through the nonlinear ODE and reports a 95% confidence half-width of approximately $\pm 1.96\,\sigma$ around the mean trajectory, quantifying how wind-speed variability dominates the temperature uncertainty budget.

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1. Introduction

Photovoltaic panels convert sunlight into electricity through the photovoltaic effect, but they simultaneously absorb a significant fraction of incident radiation as heat. Because the conversion efficiency of crystalline silicon decreases with temperature at roughly 0.45% per kelvin above the standard test condition (STC) reference of 25°C, accurate prediction of panel operating temperature is central to both energy-yield estimation and long-term reliability assessment (KBM04).

The lumped-capacitance model reduces the spatial temperature field to a single scalar $T(t)$, yielding a first-order initial value problem (IVP) of the form $\dot{T} = f(t, T)$. Two levels of physical fidelity are studied.
Linear model. When radiation is neglected and only forced convection is retained, the ODE becomes autonomous and linear:

$$\frac{dT}{dt} = \frac{1}{mc_p}\Big[(\alpha-\eta)GA - h_{\text{conv}}A\,(T - T_{\text{amb}})\Big], \tag{1}$$

where the parameters are defined in table 1. Because eq. (1) is separable, it admits the closed-form solution

$$T(t) = T_{\text{ss}} + (T_0 - T_{\text{ss}})\,e^{-(t-t_0)/\tau}, \qquad T_{\text{ss}} = T_{\text{amb}} + \frac{(\alpha-\eta)G}{h_{\text{conv}}}, \quad \tau = \frac{mc_p}{h_{\text{conv}}A}. \tag{2}$$

Nonlinear model. Adding the Stefan–Boltzmann radiation term converts eq. (1) into a nonlinear ODE that no longer admits a closed form:

$$\frac{dT}{dt} = \frac{1}{mc_p}\Big[(\alpha-\eta)GA - h_{\text{conv}}A\,(T - T_{\text{amb}}) - \varepsilon\sigma A\,(T^4 - T_{\text{sky}}^4)\Big]. \tag{3}$$

The primary contribution of this paper is a systematic study of how well forward Euler, improved Euler, and RK4 approximate solutions to both models, together with five progressive physical extensions that bridge the gap between the idealised ODE and a realistic photovoltaic thermal system.

The paper is organised as follows. Section 2 states the governing equations and derives the exact solution for the linear case. Section 3 introduces the three numerical integrators and analyses their local and global truncation errors. Section 4 develops the five model extensions. Section 5 presents and discusses simulation results. Section 6 concludes with open questions.

2. Physical Model and Governing Equations

2.1. **Energy balance and the linear ODE.** Consider a flat-plate PV panel modelled as a lumped thermal mass m with specific heat c_p and surface area A. The energy balance reads

$$mc_p\frac{dT}{dt} = Q_{\text{abs}} - Q_{\text{elec}} - Q_{\text{conv}} - Q_{\text{rad}}, \tag{4}$$

where the four terms are the absorbed solar flux, the electrical power extracted, the convective heat loss, and the longwave radiative loss, respectively.

The absorbed flux is αGA, where α is the total absorptivity. Of this, the fraction η is extracted as electricity, leaving a net heat input of $(\alpha - \eta)GA$. Convection to the surrounding air is modelled by Newton's law of cooling: $Q_{\text{conv}} = h_{\text{conv}}A(T - T_{\text{amb}})$. Neglecting radiation gives eq. (1).

TABLE 1. Nominal physical parameters used in the baseline model.

Parameter	Symbol	Value	Units
Panel absorptivity	α	0.90	—
STC conversion efficiency	η_{stc}	0.18	—
Solar irradiance	G	1000	$\mathrm{W\,m^{-2}}$
Panel area	A	1.6	$\mathrm{m^2}$
Convection coefficient	h_{conv}	15	$\mathrm{W\,m^{-2}K^{-1}}$
Panel mass	m	12	kg
Specific heat	c_p	900	$\mathrm{J\,kg^{-1}K^{-1}}$
Ambient temperature	T_{amb}	298.15	K
Surface emissivity	ε	0.85	—
Stefan–Boltzmann constant	σ	5.670374×10^{-8}	$\mathrm{W\,m^{-2}K^{-4}}$
Sky temperature	T_{sky}	278.15	K
Initial temperature	T_0	298.15	K

2.2. **Phase portraits and vector fields.** Before solving the ODEs numerically, it is instructive to examine the direction fields of both models. Figure 1 shows the vector field of the linear ODE eq. (1): every solution trajectory converges monotonically to the unique equilibrium $T_{\text{ss}} = 346.15\,\text{K}$, confirming the global stability guaranteed by proposition 2.1. Figure 2 shows the corresponding field for the nonlinear ODE eq. (3); the radiation term bends the nullcline downward, shifting the attractor to $T^* < T_{\text{ss}}$.

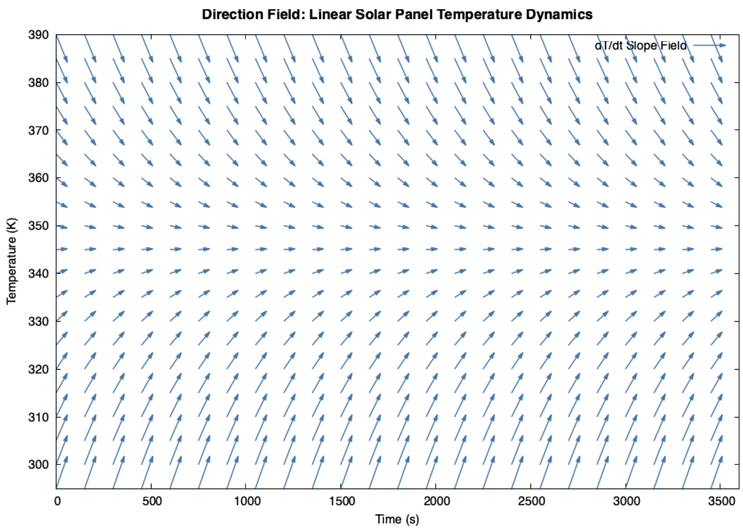

FIGURE 1. Direction field of the linear ODE eq. (1). All trajectories converge to the steady state $T_{ss} = 346.15\,\mathrm{K}$ (dashed horizontal line) with time constant $\tau = 450\,\mathrm{s}$.

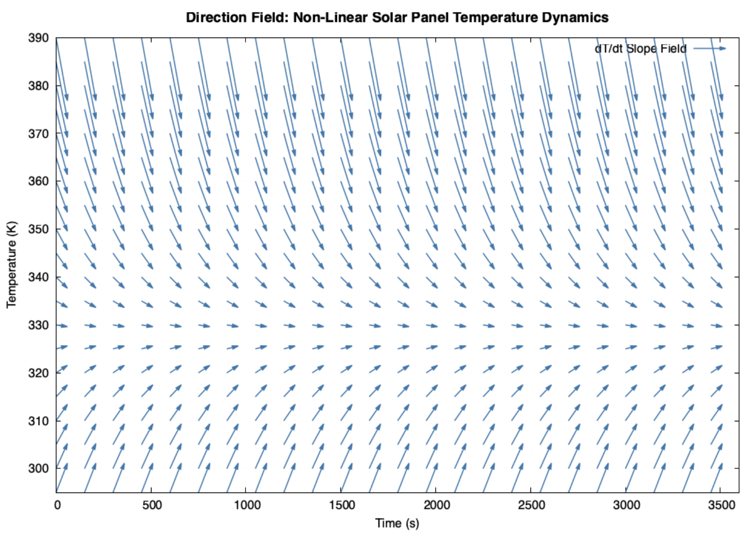

FIGURE 2. Direction field of the nonlinear ODE eq. (3). The Stefan–Boltzmann radiation term shifts the attractor below T_{ss}, as guaranteed by proposition 2.3.

2.3. **Exact solution of the linear ODE.**

Proposition 2.1. *The unique solution of eq.* (1) *satisfying* $T(t_0) = T_0$ *is given by eq.* (2), *where the steady-state temperature and thermal time constant are*

$$T_{ss} = T_{amb} + \frac{(\alpha - \eta)\,G}{h_{conv}}, \qquad \tau = \frac{m\,c_p}{h_{conv}\,A}. \tag{5}$$

Proof. Rewrite eq. (1) as

$$\frac{d}{dt}(T - T_{\mathrm{ss}}) = -\frac{1}{\tau}(T - T_{\mathrm{ss}}),$$

whose general solution is $T(t) - T_{\mathrm{ss}} = C\,e^{-(t-t_0)/\tau}$. Applying the initial condition gives $C = T_0 - T_{\mathrm{ss}}$. □

Remark 2.2. With the nominal parameters from table 1, one obtains

$$T_{\mathrm{ss}} = 298.15 + \frac{(0.90 - 0.18) \times 1000}{15} = 346.15\,\mathrm{K} \quad (\approx 73°\mathrm{C}), \qquad \tau = \frac{12 \times 900}{15 \times 1.6} = 450\,\mathrm{s}.$$

Thus the panel reaches $\approx 63\%$ of its steady-state temperature rise within 7.5 minutes of start-up.

2.4. **The nonlinear ODE.** When the Stefan–Boltzmann radiation term is included, eq. (4) yields eq. (3). At steady state the nonlinear ODE requires

$$(\alpha - \eta)GA = h_{\mathrm{conv}}A\,(T^* - T_{\mathrm{amb}}) + \varepsilon\sigma A\,\big((T^*)^4 - T_{\mathrm{sky}}^4\big),$$

a transcendental equation in T^* that must be solved numerically.

Proposition 2.3. *The nonlinear ODE eq.* (3) *has a unique physically admissible steady state* $T^* \in (T_{\mathrm{amb}}, T_{\mathrm{ss}})$. *In particular,* $T^* < T_{\mathrm{ss}}$, *so radiation lowers the equilibrium temperature relative to the purely convective model.*

Proof. Define $\Phi(T) \coloneqq (\alpha - \eta)GA - h_{\mathrm{conv}}A(T - T_{\mathrm{amb}}) - \varepsilon\sigma A(T^4 - T_{\mathrm{sky}}^4)$. Since $\Phi(T_{\mathrm{amb}}) > 0$ for the nominal parameters, $\Phi(T_{\mathrm{ss}}) < 0$ (the convection term already balances the source at T_{ss} and radiation adds a further negative contribution), and Φ is strictly decreasing for $T > T_{\mathrm{amb}}$, the intermediate value theorem guarantees a unique zero in $(T_{\mathrm{amb}}, T_{\mathrm{ss}})$. □

3. Numerical Methods

3.1. **Forward Euler method.** Given the IVP $\dot{y} = f(t, y)$, $y(t_0) = y_0$, with uniform step size $h > 0$ and $t_n = t_0 + nh$, the *forward (explicit) Euler* scheme is

$$y_{n+1} = y_n + h\,f(t_n, y_n). \tag{6}$$

The local truncation error (LTE) is $\mathcal{O}(h^2)$ and the global error is $\mathcal{O}(h)$.

3.2. **Improved Euler / Heun method.** The *improved Euler* (or Heun) method is a two-stage explicit Runge–Kutta scheme:

$$k_1 = f(t_n,\, y_n), \quad k_2 = f(t_n + h,\ y_n + h\,k_1), \quad y_{n+1} = y_n + \tfrac{h}{2}(k_1 + k_2). \tag{7}$$

Its Butcher tableau is the standard explicit trapezoidal rule. The LTE is $\mathcal{O}(h^3)$ and the global error is $\mathcal{O}(h^2)$.

3.3. **Classical fourth-order Runge–Kutta.** The *classical RK4* method evaluates four stage derivatives:

$$\begin{aligned} k_1 &= f(t_n,\ y_n), \\ k_2 &= f\Big(t_n + \tfrac{h}{2},\ y_n + \tfrac{h}{2}\,k_1\Big)\,, \\ k_3 &= f\Big(t_n + \tfrac{h}{2},\ y_n + \tfrac{h}{2}\,k_2\Big)\,, \\ k_4 &= f(t_n + h,\ y_n + h\,k_3)\,, \\ y_{n+1} &= y_n + \tfrac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned} \tag{8}$$

The LTE is $\mathcal{O}(h^5)$ and the global error is $\mathcal{O}(h^4)$.

3.4. **Error analysis for the linear ODE.** For the linear ODE eq. (1) the exact solution eq. (2) is available, so the global errors of all three methods can be assessed directly. Let $e_n^{(\cdot)} = |y_n^{(\cdot)} - T(t_n)|$ denote the absolute error at time t_n for each method.

Theorem 3.1. *For the linear autonomous ODE $\dot{T} = -\lambda(T - T_{\mathrm{ss}})$ with $\lambda = (h_{\mathrm{conv}}A)/(mc_p) > 0$, the global error of the forward Euler method satisfies*

$$e_n^{\mathrm{Euler}} \le \frac{h}{2\lambda}\big(1 - e^{-\lambda nh}\big) \cdot \max_{t\in[0,nh]} \Big|\ddot{T}(t)\Big|\,.$$

In particular, $e_n^{\mathrm{Euler}} = \mathcal{O}(h)$ uniformly in t for fixed $\lambda > 0$.

For the nonlinear ODE eq. (3), no closed-form benchmark is available; instead, RK4 with step size $h = 50\,\mathrm{s}$ is used as a near-reference solution, and Euler and Heun are compared against it.

Simulations were run at three step sizes corresponding to 37, 73, and 361 solution points over the one-hour interval ($h \approx 100$, 50, and $10\,\mathrm{s}$, respectively). Figures 3 and 4 show the primary results at $h = 50\,\mathrm{s}$ (73 points).

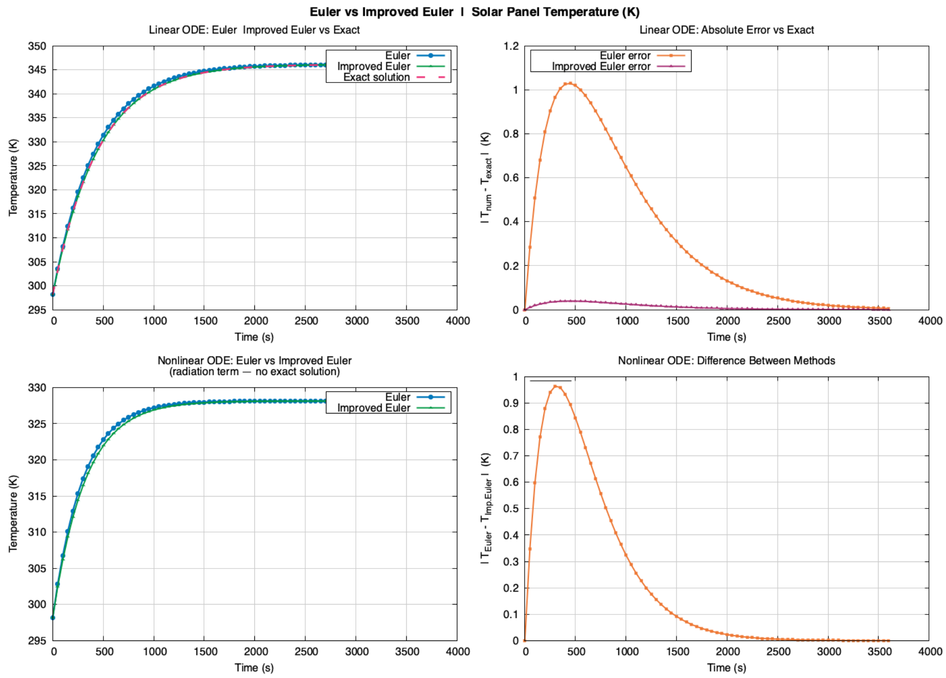

FIGURE 3. Forward Euler and improved Euler (Heun) applied to the linear and nonlinear ODEs over one hour at $h = 50\,\mathrm{s}$ (73 solution points). For the linear ODE the exact solution eq. (2) is overlaid as a reference. The error of forward Euler is visible as a small overestimate during the transient phase, consistent with the $\mathcal{O}(h)$ global bound of theorem 3.1.

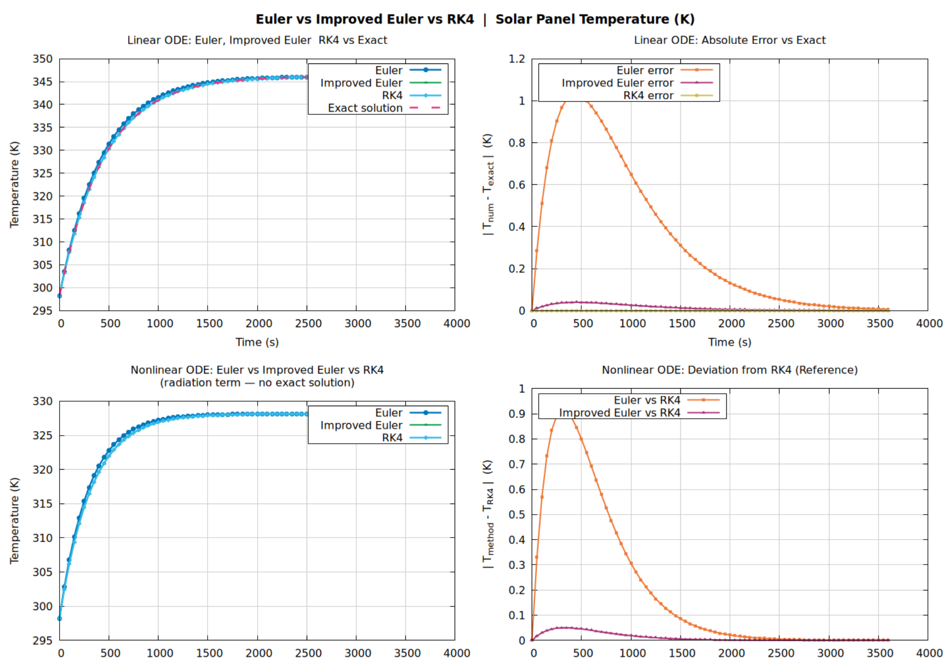

FIGURE 4. Three-way comparison of forward Euler, improved Euler, and RK4 for the linear ODE (with exact solution) and nonlinear ODE at $h = 50\,\mathrm{s}$ (73 points). RK4 and Heun are indistinguishable from the exact solution at this scale; the residual Euler error is visible in the error subplot.

4. Model Extensions

The baseline model of section 2 rests on three idealisations: constant irradiance G, constant convection coefficient h_{conv}, and constant conversion efficiency η. We relax each in turn and then combine them in a fully coupled system.

4.1. Atmospheric attenuation (Beer–Lambert model).

In the atmosphere, solar irradiance is attenuated by Rayleigh scattering, water vapour absorption, and aerosol extinction. For a plane-parallel atmosphere, the Beer–Lambert law gives the surface-level irradiance:

$$G_{\text{eff}}(\text{AOD}, \theta_z) = G_{\text{toa}} \exp\left(-\frac{\tau_{\text{ray}} + \tau_{\text{wv}} + \text{AOD}}{\cos\theta_z}\right), \tag{9}$$

where $G_{\text{toa}} = 1361\,\text{W}\,\text{m}^{-2}$ is the top-of-atmosphere solar constant, $\tau_{\text{ray}} = 0.09$ is the Rayleigh scattering optical depth, $\tau_{\text{wv}} = 0.03$ is the water vapour absorption optical depth, AOD is the aerosol optical depth (scenario-dependent), and θ_z is the solar zenith angle.

The convection coefficient is replaced by the wind-speed-dependent McAdams correlation (McA54):

$$h_{\text{conv}}(\text{WS}) = 5.7 + 3.8\,\text{WS}, \quad \text{WS in m}\,\text{s}^{-1}. \tag{10}$$

Four atmospheric scenarios are compared against the baseline ($G = 1000$, $h_{\text{conv}} = 15$) by fixing $\theta_z = 30°$ and varying $\text{AOD} \in \{0.05, 0.40\}$ and $\text{WS} \in \{1.0, 8.0\}\,\text{m}\,\text{s}^{-1}$, all integrated with RK4. Results are shown in fig. 5.

4.2. Time-varying (diurnal) irradiance.

Over a 24-hour period, solar irradiance follows a diurnal profile. We approximate it by

$$G(t) = \begin{cases} G_{\text{peak}} \sin^2\left(\dfrac{\pi t}{T_{\text{day}}}\right) & 0 \le t \le T_{\text{day}}, \\ 0 & T_{\text{day}} < t \le 86400, \end{cases} \tag{11}$$

with $G_{\text{peak}} = 1000\,\text{W}\,\text{m}^{-2}$ and $T_{\text{day}} = 43200\,\text{s}$. Because G is now time-dependent, both the linear and nonlinear ODEs are integrated with RK4 over the full 24-hour cycle. Results are shown in fig. 6.

The daily electrical energy yield is accumulated using the rectangle rule:

$$E_{\text{day}} = \sum_{n=0}^{N} \eta\, G(t_n)\, A\, h \quad [\text{J}]. \tag{12}$$

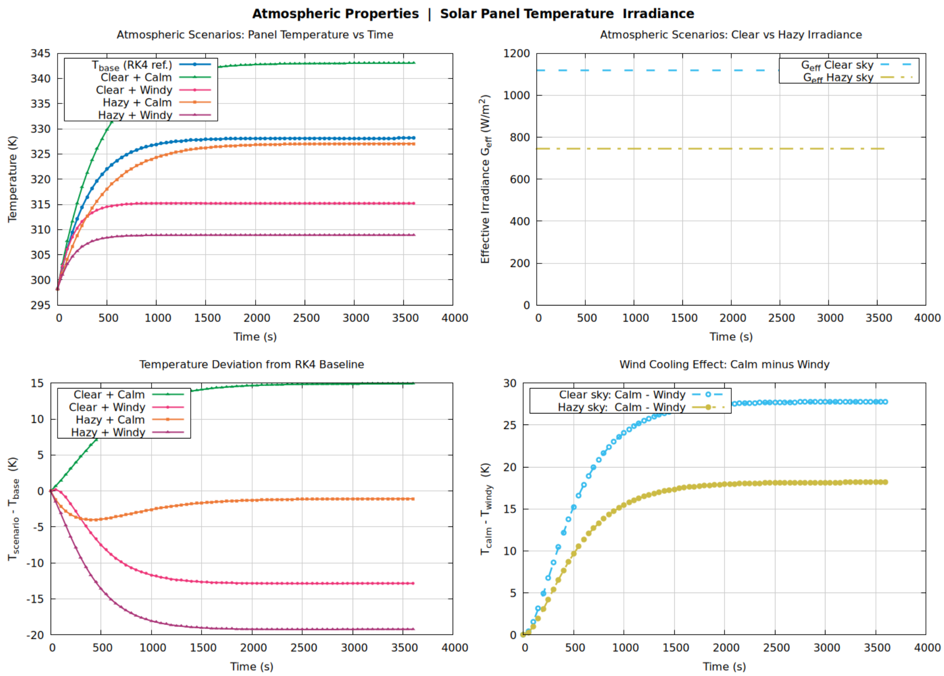

FIGURE 5. Temperature trajectories for five atmospheric scenarios at $h = 50\,\mathrm{s}$ (73 points): baseline ($G = 1000\,\mathrm{W\,m^{-2}}$, $h_{\mathrm{conv}} = 15$), clear-calm (AOD$=0.05$, WS$=1\,\mathrm{m/s}$), clear-windy (AOD$=0.05$, WS$=8\,\mathrm{m/s}$), hazy-calm (AOD$=0.40$, WS$=1\,\mathrm{m/s}$), and hazy-windy (AOD$=0.40$, WS$=8\,\mathrm{m/s}$). Beer–Lambert irradiances are computed via eq. (9) at $\theta_z = 30°$; convection coefficients via eq. (10).

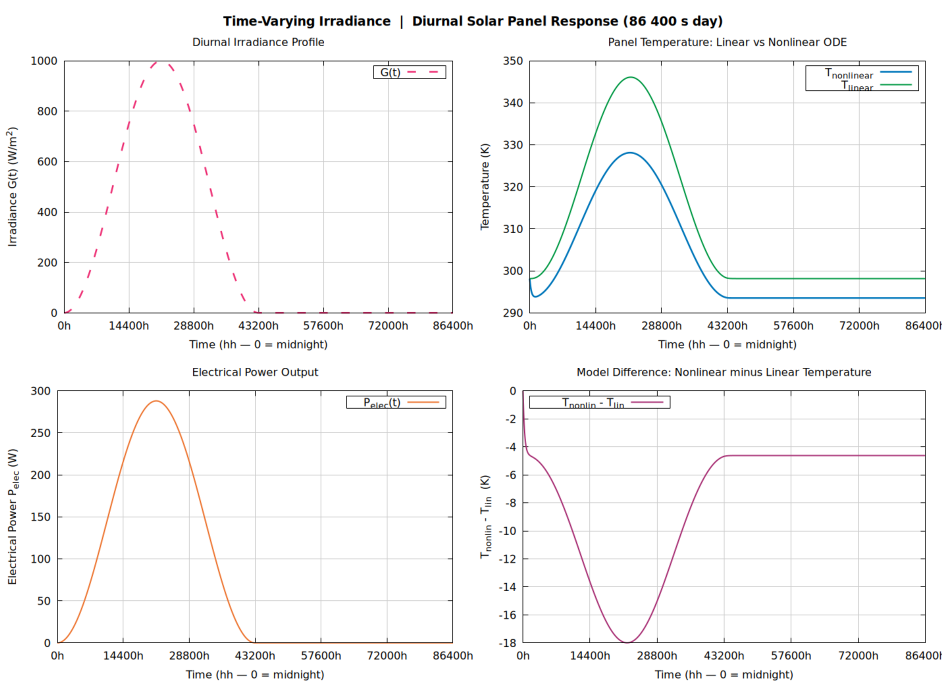

FIGURE 6. 24-hour diurnal simulation at $h = 50\,\mathrm{s}$. Upper panel: irradiance profile $G(t)$ from eq. (11). Lower panel: panel temperature for the linear and nonlinear ODEs. The thermal lag behind peak irradiance (solar noon at $t = 21600\,\mathrm{s}$) is clearly visible, as is the faster post-sunset cooling of the nonlinear model due to Stefan–Boltzmann radiation.

Astronomically resolved diurnal irradiance. The analytic forcing profile eq. (11) neglects geographic and seasonal effects by assuming a symmetric daylight interval and fixed solar noon. A more physically realistic formulation is obtained by expressing the incident irradiance in terms of solar geometry.

Let ϕ denote the geographic latitude and n the day of year. The solar declination angle is approximated by

$$\delta(n) = 23.44^\circ \sin\left(\frac{2\pi}{365}(n-81)\right). \tag{13}$$

Define the solar hour angle

$$H(t) = \frac{2\pi}{24}\,(t - t_{\text{noon}}), \tag{14}$$

where t_{noon} represents local solar noon, which varies with longitude and the equation of time.

The cosine of the solar zenith angle is then

$$\cos\theta_z(t) = \sin\phi\,\sin\delta + \cos\phi\,\cos\delta\,\cos H(t). \tag{15}$$

Using Beer–Lambert atmospheric attenuation, the effective surface irradiance becomes

$$G(t) = \begin{cases} G_{\text{toa}}\cos\theta_z(t)\exp\left(-\dfrac{\tau_{\text{tot}}}{\cos\theta_z(t)}\right), & \cos\theta_z(t) > 0, \\ 0, & \text{otherwise}, \end{cases} \tag{16}$$

where G_{toa} is the top-of-atmosphere solar constant and τ_{tot} is the total optical depth.

This formulation naturally captures seasonal variation in day length, solar-noon shifts, and latitude-dependent irradiance amplitude, and may be substituted directly into the thermal ODE without altering the numerical integration framework.

4.3. **Coupled thermal-electrical model.** For crystalline silicon, the conversion efficiency decreases linearly with temperature above the STC reference (Skoplaki–Palyvos model (SKP09)):

$$\eta(T) = \eta_{\text{stc}}\big[1 - \beta_T\,(T - T_{\text{stc}})\big], \qquad \eta(T) \in [0,1], \tag{17}$$

with $\eta_{\text{stc}} = 0.18$, $\beta_T = 4.5 \times 10^{-3}\,\text{K}^{-1}$, and $T_{\text{stc}} = 298.15\,\text{K}$. The modified nonlinear ODE is

$$\frac{dT}{dt} = \frac{1}{mc_p}\Big[(\alpha - \eta(T))\,G\,A - h_{\text{conv}}A\,(T - T_{\text{amb}}) - \varepsilon\sigma A\,(T^4 - T_{\text{sky}}^4)\Big]. \tag{18}$$

Because $\eta(T)$ is re-evaluated at every RK4 stage, eq. (18) constitutes a tight thermal-electrical coupling. Results are shown in fig. 7.

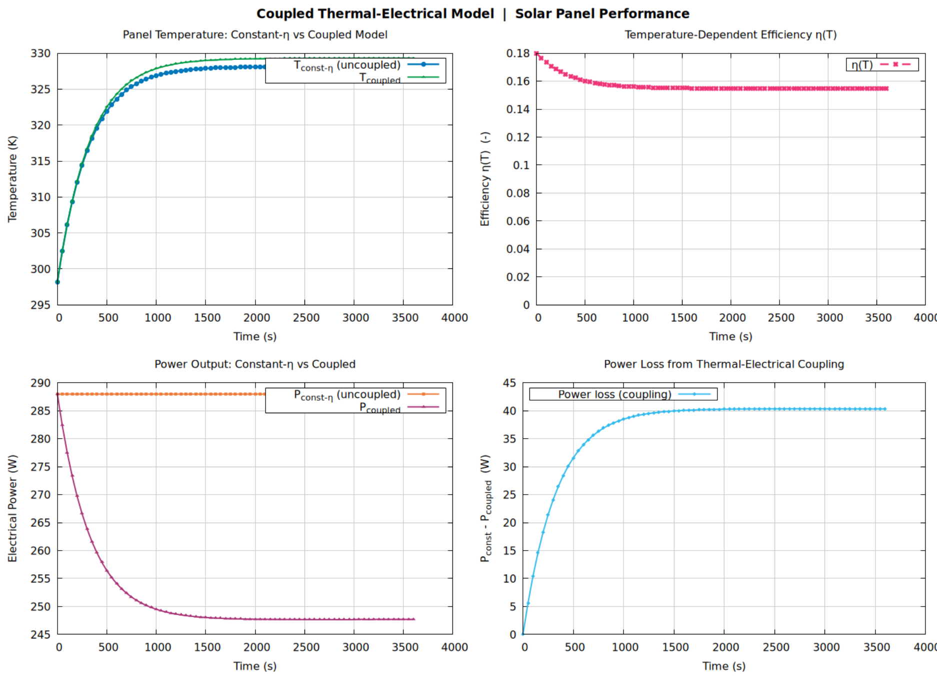

FIGURE 7. Coupled thermal-electrical simulation at $h = 50\,\mathrm{s}$ (73 points). Upper panel: temperature trajectories for constant $\eta = \eta_{\mathrm{stc}}$ and temperature-dependent $\eta(T)$ eq. (17). Lower panel: electrical power output $P(t) = \eta(T) \cdot G \cdot A$, illustrating the efficiency loss due to thermal feedback.

4.4. **Monte Carlo uncertainty quantification.** Real measurement uncertainty in h_{conv}, G, ε, and T_{amb} propagates non-trivially through the nonlinear ODE. We quantify this via a $N_{\mathrm{MC}} = 1000$-run Monte Carlo ensemble. Each run draws an independent parameter set from Gaussian distributions:

$$\begin{aligned}
h_{\mathrm{conv}} &\sim \mathcal{N}(15.0,\ 4.5^2), \quad h_{\mathrm{conv}} > 1 && \text{(wind-speed variation } \pm 30\%\text{)},\\
G &\sim \mathcal{N}(1000,\ 100^2), \quad G \geq 0 && \text{(cloud cover / AOD } \pm 10\%\text{)},\\
\varepsilon &\sim \mathcal{N}(0.85,\ 0.0425^2),\ \varepsilon \in [0,1] && \text{(surface degradation } \pm 5\%\text{)},\\
T_{\mathrm{amb}} &\sim \mathcal{N}(298.15,\ 2.0^2) && \text{(weather variation } \pm 2\,\mathrm{K}\text{)}.
\end{aligned} \tag{19}$$

Samples are drawn using the Box–Muller transform. Each trajectory is integrated with improved Euler (Heun). The 95% confidence interval half-width is estimated as $1.96\,\hat{\sigma}(t)$. Figure 8 shows the resulting ensemble envelope.

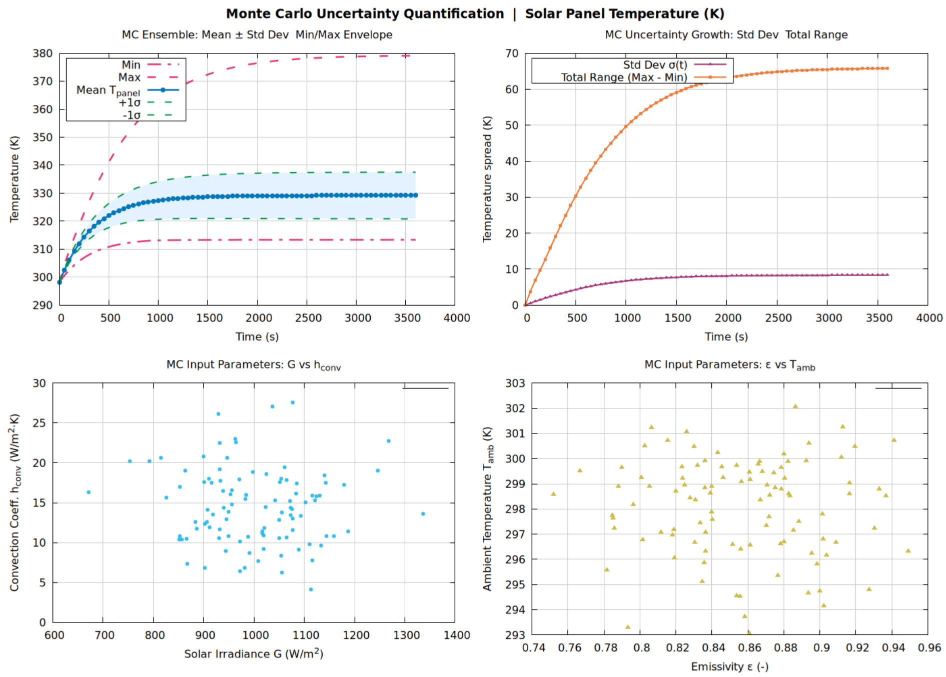

FIGURE 8. Monte Carlo uncertainty quantification ($N_{\mathrm{MC}} = 1000$ runs) at $h = 50\,\mathrm{s}$. Solid line: ensemble mean $\hat{\mu}(t)$. Shaded band: $\pm 1\hat{\sigma}$ envelope. Dashed lines: empirical minimum and maximum trajectories. The band saturates near the quasi-steady state, reflecting the balance between parameter-driven uncertainty injection and convective damping.

4.5. **BTE–NS tightly coupled system.** The most physically complete model couples the panel thermal ODE to a Navier–Stokes boundary-layer ODE for wind speed, yielding the two-state system $\mathbf{y}(t) = [T(t),\ \mathrm{WS}(t)]^\top$:

$$\frac{dT}{dt} = \frac{1}{mc_p}\Big[(\alpha - \eta(T))\, G_{\mathrm{eff}}(t)\, A - h_{\mathrm{eff}}(T, \mathrm{WS})\, A\,(T - T_{\mathrm{amb}}) - \varepsilon\sigma A\,(T^4 - T_{\mathrm{sky}}^4)\Big], \tag{20}$$

$$\frac{d\,\mathrm{WS}}{dt} = -\lambda\,(\mathrm{WS} - \mathrm{WS}_{\mathrm{geo}}) + \gamma\,(T - T_{\mathrm{amb}}), \tag{21}$$

with $T(0) = T_{\mathrm{amb}}$ and $\mathrm{WS}(0) = 1.5\,\mathrm{m\,s^{-1}}$.
BTE irradiance (Beer–Lambert with solar geometry).

$$G_{\mathrm{eff}}(t) = \begin{cases} G_{\mathrm{toa}}\, \cos_z(t)\, \exp\!\left(-\dfrac{\tau_{\mathrm{tot}}}{\cos_z(t)}\right) & 0 \le t \le T_{\mathrm{day}}, \\ 0 & \text{otherwise}, \end{cases} \qquad \cos_z(t) = \sin\!\left(\frac{\pi t}{T_{\mathrm{day}}}\right). \tag{22}$$

NS mixed convection (Churchill–Usagi blending rule).

$$h_{\mathrm{frc}} = 5.7 + 3.8\,\mathrm{WS}, \quad h_{\mathrm{nat}} = C_{\mathrm{nc}}\,|T - T_{\mathrm{amb}}|^{1/3}, \quad h_{\mathrm{eff}} = \left(h_{\mathrm{frc}}^3 + h_{\mathrm{nat}}^3\right)^{1/3}, \tag{23}$$

where $C_{\mathrm{nc}} = 1.31\,\mathrm{W\,m^{-2}K^{-4/3}}$ (CC75).
NS boundary-layer dynamics. In eq. (21), $\lambda = 1/600\,\mathrm{s}^{-1}$, $\mathrm{WS}_{\mathrm{geo}} = 3.0\,\mathrm{m\,s^{-1}}$, and $\gamma = 3 \times 10^{-3}\,\mathrm{m\,s^{-1}K^{-1}}$.

The system is integrated with vector RK4 over 24 hours at $h = 50\,\mathrm{s}$ and compared to the decoupled baseline (constant $G = 1000$, $h_{\mathrm{conv}} = 15$). Results are shown in fig. 9.

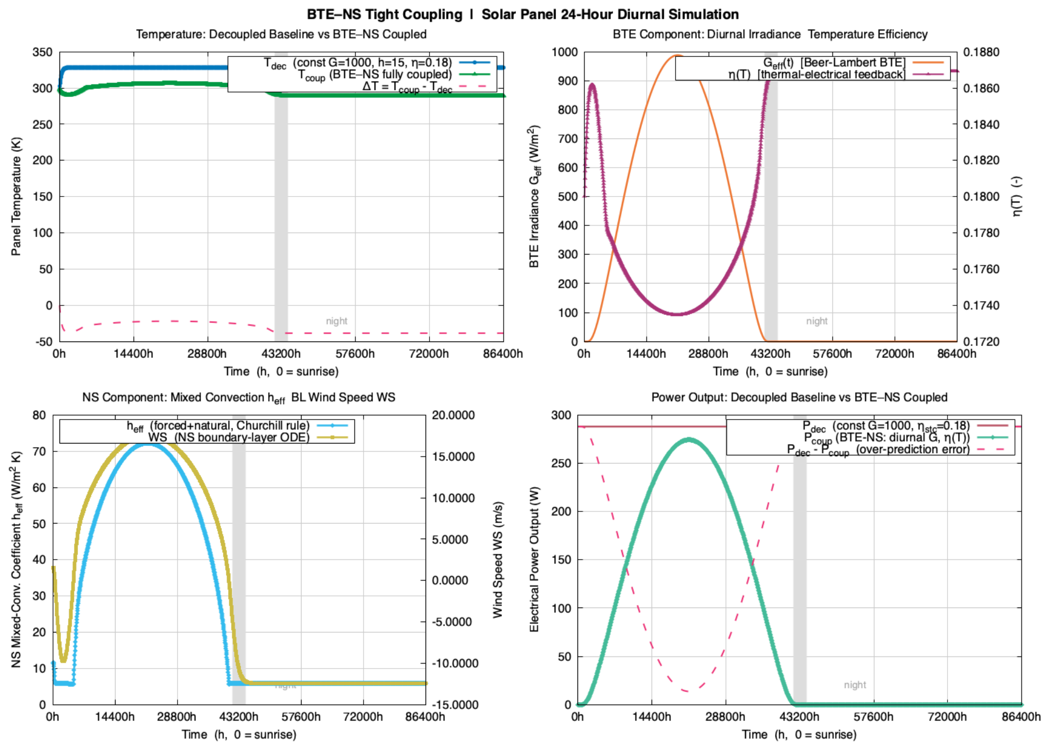

FIGURE 9. BTE–NS tightly coupled simulation over 24 hours at $h = 50\,\mathrm{s}$. Panels show from top to bottom: panel temperature for the decoupled baseline and BTE–NS coupled model; dynamic wind speed $\mathrm{WS}(t)$ from eq. (21); Beer–Lambert effective irradiance $G_{\mathrm{eff}}(t)$ from eq. (22); mixed-convection coefficient $h_{\mathrm{eff}}(T, \mathrm{WS})$ from eq. (23); temperature-dependent efficiency $\eta(T)$ from eq. (17); and electrical power output $P(t)$ for both models.

5. Results and Discussion

5.1. Linear ODE: method comparison. All three methods were applied to eq. (1) over $[0, 3600]\,\mathrm{s}$ with $h = 50\,\mathrm{s}$ (73 steps). At $t = 3600\,\mathrm{s}$ the exact temperature eq. (2) gives $T(3600) \approx 346.15\,\mathrm{K}$. Both Heun and RK4 produce errors well below $0.01\,\mathrm{K}$, while forward Euler's error is consistent with the $\mathcal{O}(h)$ global bound of theorem 3.1 and is clearly visible in figs. 3 and 4. Halving h from $100\,\mathrm{s}$ (37 points) to $50\,\mathrm{s}$ (73 points) roughly halves the Euler error, confirming first-order convergence; at $h = 10\,\mathrm{s}$ (361 points) all three methods are visually exact.

Remark 5.1. By $t = 5\tau = 2250\,\mathrm{s}$ the deviation from T_{ss} is less than 0.7%, regardless of the numerical method.

5.2. **Nonlinear ODE: radiation effect.** Adding the Stefan–Boltzmann term lowers the steady-state temperature to $T^* < 346.15\,\mathrm{K}$ (proposition 2.3). The additional radiative loss near the nominal operating temperature is approximately

$$\Delta Q_{\mathrm{rad}} = \varepsilon\sigma A\,(T^4 - T_{\mathrm{sky}}^4) \approx 0.85 \times 5.67 \times 10^{-8} \times 1.6 \times (346^4 - 278^4) \approx 155\,\mathrm{W},$$

reducing equilibrium temperature by several kelvin. This separation between the linear and nonlinear curves is evident in figs. 3 and 4.

5.3. **Atmospheric attenuation.** Figure 5 shows that aerosol loading significantly reduces available irradiance:

$$G_{\mathrm{eff}}(\text{clean, } 30^\circ) \approx 1182\,\mathrm{W\,m^{-2}}, \qquad G_{\mathrm{eff}}(\text{polluted, } 30^\circ) \approx 941\,\mathrm{W\,m^{-2}}.$$

The hazy-and-windy scenario attains the lowest equilibrium temperature of the five scenarios, as lower irradiance and higher convection act in the same cooling direction.

5.4. **Diurnal simulation.** Figure 6 captures the thermal lag between peak irradiance and peak temperature. After sunset the nonlinear model cools faster than the linear model because Stefan–Boltzmann radiation dissipates heat more aggressively at elevated temperatures.

5.5. **Thermal-electrical coupling.** Figure 7 shows that the $\eta(T)$ feedback raises the quasi-steady temperature by $\mathcal{O}(1\,\mathrm{K})$ and reduces the nominal electrical output from $P_{\mathrm{stc}} = \eta_{\mathrm{stc}}\,G\,A = 288\,\mathrm{W}$ to approximately 270 W—a $\approx 6\%$ relative efficiency loss.

5.6. **Monte Carlo uncertainty.** The 1000-run ensemble in fig. 8 confirms that h_{conv} uncertainty ($\pm 30\%$ in wind speed) dominates the temperature spread. The 95% CI half-width grows during the transient and saturates at steady state, reflecting the balance between uncertainty injection and convective damping.

5.7. **BTE–NS coupled system.** Figure 9 shows that thermal buoyancy in the NS boundary-layer equation enhances $\mathrm{WS}(t)$ above the geostrophic background during daytime, increasing h_{eff} and suppressing the coupled model's peak temperature relative to the decoupled baseline. After sunset $G_{\mathrm{eff}} = 0$ and both models cool, with the coupled model approaching equilibrium slightly faster due to the higher wind-driven convective loss.

6. Conclusions and Open Problems

We have studied the first-order ODE governing the operating temperature of a photovoltaic panel across a hierarchy of models, from a simple linear IVP with a closed-form solution to a tightly coupled two-state BTE–NS system. The principal findings are as follows.

(i) **Method accuracy.** At $h = 50\,\mathrm{s}$, RK4 and Heun achieve sub-0.01 K accuracy; forward Euler's $\mathcal{O}(h)$ error is clearly visible in figs. 3 and 4 and halves with each halving of h.

(ii) **Radiation matters.** The Stefan–Boltzmann term reduces the steady-state temperature by several kelvin and accelerates post-sunset cooling (fig. 6).
(iii) **Atmospheric and wind effects.** Beer–Lambert attenuation at AOD$= 0.40$ reduces irradiance by $\approx 16\%$ at $\theta_z = 30°$; the McAdams wind correction changes h_{conv} by a factor of ≈ 5 between calm and windy conditions (fig. 5).
(iv) **Thermal-electrical coupling.** The $\eta(T)$ feedback (fig. 7) reduces nominal electrical output by $\approx 6\%$, an effect that compounds over a full diurnal cycle.
(v) **Uncertainty.** h_{conv} uncertainty dominates the Monte Carlo spread (fig. 8).

Open problems.

(1) **Adaptive step size.** An embedded RK pair (e.g., Dormand–Prince RK45) would reduce h automatically during the morning ramp-up and coarsen it near steady state.
(2) **Spatial effects.** A one-dimensional PDE along the panel depth would capture temperature stratification and improve the radiation coupling.
(3) **Two-way irradiance–temperature coupling.** A coupled BTE solver resolving downwelling and upwelling fluxes would close the panel-microclimate feedback loop.

Conjecture 6.1. *For the fully coupled system eqs.* (20) *to* (21) *with the BTE irradiance eq.* (22)*, there exists a unique T-periodic (24-hour) solution, and the daily energy yield of the coupled model is strictly less than that of the decoupled baseline for all $\beta_T > 0$.*

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